

Conference call

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Monday 5 October 2026



Join Slack

[accelerating-astrophysical-
discovery.github.io/members/](https://accelerating-astrophysical-discovery.github.io/members/)

accelerating-astrophysical-discovery.github.io

Goals of AAD consortium

PILLAR 1

Formalising science

Make hypotheses, evidence and their relationships explicit.

PILLAR 2

An open-science model

Develop multimodal reasoning to discover new links within the formalism.

119

People

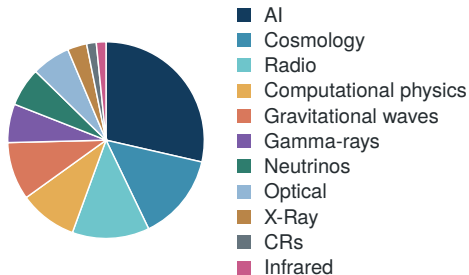
68

Institutes

13

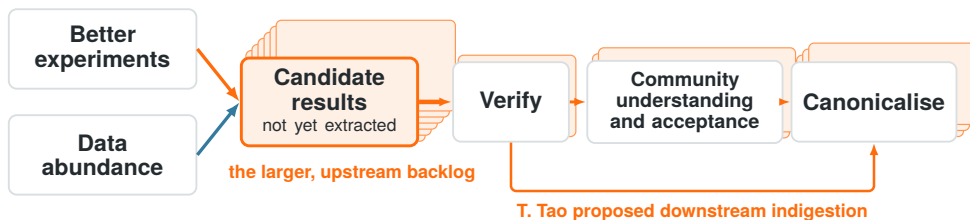
Countries

CONSORTIUM FIELDS



Our postulate

High-quality observations accumulate faster than humanity can unlock their insights



We have two bottlenecks:

1. Observations outpace our capacity to extract understanding.
2. AI-enhanced results outpace the community's capacity to digest them.

Workshop outcomes and follow-up

PRISM — Argonne National Laboratory

An initial prototype fusing multimodal data and language reasoning with simulation feedback.

A preliminary dataset inventory

Candidate observations, simulations and linked data for training and evaluation.

A preliminary formalism of science

Explicit links among systems, theories, models, observations, simulations and discussion.

We performed a first successful demonstration with the MMU dataset.

Two connected technical workgroups

Formalising science and data

Develop the scientific concepts, relationships and evidence rules; curate aligned multimodal datasets with provenance.

Modelling and federated training

Connect multimodal data and language; test anomaly detection, simulation/emulation and learning from simulation feedback. Bring training to distributed datasets across institutions.

Conversations since the workshop

1 **Terence Tao**

Offered a guest spot on his blog.

2 **Bryna Kra**

The mathematics community's activities and process: inspiration for inclusive drafting and collective review.

3 **Google**

Discussions about possible support and collaboration.

4 **Discussions with adjacent fields**

Discussions with researchers already working on scientific formalisation.

Why we need to move now

AI-enabled science is taking shape now.

Industry is moving quickly

Formalising and automating research workflows, connecting models to scientific data and tools.

Our risk: potentially unhealthy outcomes for our science

If the scientific community is absent, systems may be built around priorities that do not serve our science.

Our collaborative process

1. Use our time together to listen to feedback and objections.
2. Divide and conquer work into concrete artifacts for review.
3. Detailed summaries are prepared after each meeting and made available.

Our declaration: two bottlenecks, three propositions

24 September draft

The discovery backlog

Data accumulates faster than we can extract insight.

Scientific indigestion

Results accumulate faster than we can verify, understand and teach them.

- 1 Human intuition must guide discovery**
Consequential human participation, beyond final verification.
- 2 Scientific knowledge must be formally maintained**
Explicit, traceable links among theories, models, data and evidence.
- 3 Open science must own the AI that is used to accelerate discovery**
Openly maintained models and infrastructure.

Collective feedback

1. We should use measured language and make the reasoning clear.
2. Strong claims should be supported by evidence and concrete examples.
3. We should ground human participation in scientific quality and learning.
4. We should prioritise scientific understanding, not just more outputs.
5. Formal knowledge maintenance should not prescribe a particular implementation.
6. We should reconsider categorical ownership requirements and the role of commercial AI.

Live declaration document